

Action, Trajectories and Gaps: A Research Programme

navstokgap research working notes

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Abstract

A restartable programme for testing action-scale restrictions and gap formation in elementary mechanics. It separates classical obstructions, conditional quantum bounds, reconstruction hypotheses and future field-theory comparisons. This is a research design, not a claim that quantum mechanics or a mass gap has been derived. The maintained source is `research/PROGRAMME.md`; this LaTeX is generated from it. Operational task state is kept separately in the repository.

1 Research programme: action, trajectories and gaps

Version 1, 2026-09-05. This is a multi-session research programme, not a claimed solution or a promise that the desired conclusion is true.

1.1 Aim and possible outcomes

Understand whether a precisely stated extension or reinterpretation of mechanical trajectories explains a nonzero action scale, and which Lagrangians, state spaces and boundary conditions produce an actual lower bound. The originating heuristic relates Newton’s vanishing geometric areas to action differences. Two admissible positive outcomes are:

1. A reconstruction/formalism for quantum mechanics, with its independent physical assumptions exposed and its relationship to existing reconstructions established.
2. A rigorous toy-model account of gap formation, identifying the quantity bounded and the hypotheses responsible for the bound.

These outcomes can coexist. A rigorous obstruction, counterexample or demonstration that a proposed reconstruction assumes the desired conclusion is also a result. Novelty requires a literature comparison and ultimately specialist review.

1.2 Questions that must not be conflated

Target	Precise question	Current status
Q0: classical existence	For which initial data, force regularity and collision conventions does an ODE trajectory exist uniquely and globally?	Constant force solved; central singularities not audited
Q1: action-value gap	Is the infimum of positive same-endpoint action differences over a specified path class positive?	No for the unrestricted constant-force variation family

Target	Precise question	Current status
Q2: nonzero action scale	Can independent operational/composition premises force a parameter with action units to be nonzero?	Open programme; Newtonian equations alone do not force it
Q3: operational bound	With a fixed measurement protocol, resource budget and error target, what relative action is distinguishable?	Conditional two-arm threshold derived; quantum inputs explicit
Q4: spectral gap	For a specified self-adjoint operator and Hilbert space, is the spectrum separated above its ground sector?	Separate classical Hessian and quantum Hamiltonian tests
Q5: field-theory transfer	Which toy-model estimates survive removal of cutoffs, infinite volume and interactions?	Future comparison only; no Yang–Mills reduction

Always distinguish $h = 2\pi\hbar$ from a proposed new field. Write trajectory variations as η , a fluctuation operator as J , and any hypothetical field as χ . None is identified with $\hbar$ by notation. Absolute Hamilton actions change under endpoint terms; specify endpoint matching or a gauge convention.

1.3 Starting evidence and obstructions

For $V = -Fy$, fixed duration $T > 0$, perpendicular launch speed $v_0 > 0$ and mass $m > 0$, the checked chord comparison gives

$$\Delta S = \frac{F^2 T^3}{24m} = \frac{F}{2v_0} \mathcal{A}_{\text{lens}} = \frac{T\delta E}{12}.$$

Here $\delta E = \Delta K = -\Delta V$, not total-energy change or a variance. The family $\eta(t) = at(T - t)$ has $\Delta S = ma^2 T^3/6 \rightarrow 0$. Consequently Q1 cannot be proved for this unrestricted path class. Comparing off-shell paths is not comparing distinct solutions of the same initial-value problem. A reformulation must say what is restricted and why.

The quantum weight is $\exp(iS/\hbar)$, not $\exp(iL/\hbar)$ without a time integral. Equal phases modulo 2π do not impose an action gap. Investigate the proposed delta-prime mechanism through explicitly normalized kernels and distributional limits, not through visual resemblance. Jacobi fields describe linearized classical solutions, whereas arbitrary path variations need not be Jacobi fields.

1.4 Work packages and gates

Package	Deliverable and acceptance gate	Dependencies
WP0: durable foundations	Restart instructions, claim ledger, task queue, source rules and a working LaTeX/PDF build	Current milestone

Package	Deliverable and acceptance gate	Dependencies
WP1: Newton audit	Edition/folio concordance; Book I limits separated from Book III Classical Scholia; chronological support for every historical claim	Source access; no mathematical conjecture assumed
WP2: classical control cases	Short proofs for constant force, regular central potentials and Kepler collision cases; explicit action-gap obstruction	WP0; precise path topology
WP3: fluctuations and kernels	Jacobi operator/domain, conjugate-time examples, normalized free kernel and distributional limit; test delta-prime proposal	WP2; functional-analytic review
WP4: operational reconstruction	Finite-resource distinguishability theorem; then an axiom/dependency audit of reconstruction candidates	WP3; bibliography batch B01
WP5: gap laboratory	Free particle, oscillator, compact configuration space, Coulomb and selected confining potentials; explain which parameters and limits close gaps	WP2–4 as relevant; self-adjoint domains
WP6: companion limits	Nonrelativistic/relativistic comparison; viscosity, coercivity and auxiliary-flow gaps; explicit transfer failures to NS/YM	WP5; Millennium definitions remain fixed
WP7: papers and review	Human-readable proofs, reproducible checks, adversarial review, citation audit and optional selected formal certificates	An accepted result from a previous package

Packages WP1 and the bibliography parts of WP4–6 can run in parallel. Do not block an exact modern calculation on a disputed historical interpretation. Do not start an infinite-dimensional field-theory construction before the toy target is stable.

1.5 Model progression

1. Constant force on a finite interval: exact action geometry and no-gap control.
2. Harmonic oscillator: compare the time-dependent Dirichlet action Hessian with the quantum energy spacing. A positive Hessian coefficient is not a minimum nonzero value of an unnormalized quadratic form.
3. Free particle on a line versus a circle/box: expose confinement, volume and boundary-condition dependence. A finite-size gap may vanish at infinite volume.

4. Central potentials: separate local ODE existence, bound classical orbits, action-angle variables and quantum spectral claims. Kepler collision regularisation does not establish a quantum lower action bound.
5. Selected interacting or multiwell models only after the single-particle cases have defined the gap notion; tunnelling and parameter limits are explicit.

1.6 The two reconstruction branches

Branch A: quantum input admitted. Assume Hilbert states, Born probabilities and $e^{iS/\hbar}$ with $\hbar > 0$. Derive operational bounds and spectral examples. This can explain consequences of quantum mechanics, not its existence.

Branch B: quantum input to be derived. List candidate non-quantum premises (composition, continuity, reversible transformations, distinguishability, etc.). Find classical models satisfying subsets; identify precisely which premise excludes them. Compare existing reconstructions before selecting an axiom set. Establishing the form of a theory, the necessity of a nonzero scale and the numerical value of that scale are three separate obligations. Dimensionful numerical values require units and empirical calibration.

An effective action generated by coarse graining is another candidate, but the measure, degrees of freedom and scale must be specified. A symbol called an “h field” is an idea, not yet a model.

1.7 Initial reading priorities

The first bounded bibliography batch supplies three alternative reconstruction starting points: Hardy’s five axioms (2001), Chiribella–D’Ariano–Perinotti’s informational derivation (2011), and Masanes–Müller’s physical requirements (2011). Their different premises must be compared, not merged without argument. The coordinator has checked their abstracts/identifiers; no complete proof audit is claimed. Q01 should begin with Hardy and Chiribella–D’Ariano–Perinotti, using the third as a comparison, and track where classical alternatives cease to satisfy the chosen premises.

For the admitted-quantum branch, use Feynman’s primary phase formulation (1948), Wootters’s statistical-distance paper (1981), and the quantum Chernoff bound (2007). These are reading targets with different finite-sample/asymptotic scopes, not evidence for a classical action gap. The technical foundations paper supplies its elementary two-state calculation in full rather than borrowing a stronger unread theorem.

1.8 $1/c$, Navier–Stokes and Yang–Mills companions

Keep $\hbar$, c^{-1} , viscosity ν , observation time and volume as separate parameters. A relativistic speed limit alone still allows arbitrarily small trajectory variations with a strict speed margin. A nonrelativistic quantum model can have $\hbar > 0$ with $c^{-1} = 0$. These control models prevent identifying the parameters prematurely.

For every proposed bridge record source/target spaces, operators, physical versus auxiliary time, units, preserved estimate and limits. In particular a Poincare gap for a bounded-domain diffusion operator is not a proof of 3D Navier–Stokes regularity, and an oscillator gap is not a four-dimensional Yang–Mills vacuum gap. For a mass gap one must say how energy and mass are related (including c) and retain the physical Hilbert space and infinite-volume/continuum requirements.

1.9 Publication and stopping rules

The initial papers are a programme/protocol paper and a technical foundations note. They must be readable without this chat, and labeled working drafts until reviewed. Do not submit, post to arXiv, contact authors or spend money without user direction.

A work package finishes with a theorem, a counterexample, a documented conditional result, or a sharply stated unresolved obligation. Repeatedly renaming a failed gap claim is not progress. Record changes of target and their reason. See the task queue for bounded next work and the claim ledger for evidence status.